The Complex Dynamics of Self-Regulation: Analyzing State Transitions in Online Learning Environments

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Abstract. This study examines self-regulated learning in online environments through complex dynamic systems theory. Traditional methods often use static designs that overlook its dynamic nature. We collected survey data from 165 students during an online course to identify self-regulatory profiles and analyze transitions. The methodological innovation lies in applying Transition Network Analysis (TNA) to capture temporal patterns in self-regulation, revealing how students move between different regulatory states over time. Results identified five self-regulatory profiles: Hopeful, Discouraged, Realistic, Self-Confident, and Deregulators. Transition analysis showed stability in students' regulatory approaches, with many maintaining their initial states. Notably, 40.6% began as Discouraged, emphasizing initial conditions' importance in self-regulatory development. The findings demonstrate that self-regulation exhibits properties consistent with complex dynamic systems, including statistical stability patterns and preferred transition pathways. These insights enable targeted interventions by identifying which regulatory states are most stable and which transitions occur most frequently, advancing both theoretical understanding and practical applications in online education.

Keywords: Self-regulated learning; Complex dynamic systems; Online learning; Transition network analysis; Cluster analysis; State transitions; Learning analytics; Higher education.

1 Introduction

The rapid expansion of online learning environments has fundamentally transformed educational landscapes across higher education institutions worldwide. Self-regulation—the ability to monitor, control, and adapt one's learning strategies—has emerged as a crucial determinant of student success in online settings [1]. Traditional approaches to self-regulation often use static, cross-sectional designs, failing to capture its dynamic nature. Specifically, static approaches cannot reveal how students' regulatory behaviors evolve during a course, when critical transitions occur, or which regulatory patterns tend to persist versus change. This limitation prevents educators from understanding the temporal dynamics necessary for timely and targeted interventions. This study addresses this gap by applying complex dynamic systems theory to investigate how students' self-regulatory behaviors evolve in an online course, recognizing that learning is an emergent property from the interplay of cognitive,

motivational, and contextual factors across multiple timescales. Our research addresses two specific questions positioned early to guide the theoretical framework:
RQ1 What distinct self-regulatory profiles emerge from students' longitudinal survey responses throughout the course?
RQ2 How do students transition between different self-regulatory states during the course, and which transitions matter?

Complex Dynamic Systems Theory provides a powerful framework for understanding learning processes, particularly self-regulation in educational contexts. This theoretical approach views learning as an emergent property arising from the interaction of multiple interdependent components operating across different timescales [2]. Complex systems are characterized by non-linear interactions, sensitivity to initial conditions, and self-organization—properties that align well with the inherently dynamic nature of self-regulated learning [3, 4].

States and transitions are key concepts in complex systems thinking. Attractor states—stable configurations that the system maintains—explain why self-regulatory patterns persist despite intervention [5]. Initial states also significantly influence observed patterns due to sensitivity to initial conditions. Our examination of students' transitions between self-regulatory states in an online course builds on these foundations, identifying stable patterns and transition points where shifts in regulatory behavior occur [6].

2 Previous Literature

Understanding learning as a dynamic process has gained significant attention in educational research. Learning cannot be directly observed but rather manifests through the actions learners perform and how these actions unfold over time [7]. These learning events may provide more accurate learning indicators than self-descriptions or intentions [8]. Various analytical approaches have emerged to capture these dynamics, including process mining techniques that detect patterns in self-regulatory behaviors [9], sequence mining methods that reveal temporal behavioral sequences [4], and network-based approaches that model relationships between learning states [10]. While each method offers insights, they often focus on either structural or temporal aspects rather than integrating both dimensions. These data-driven analytical techniques focus on the actual learning processes, rather than relying on retrospective reports or static cross-sectional data [11].

Transition Network Analysis (TNA) addresses this limitation by modeling the both temporal sequences and relational structures. TNA captures transition probabilities between states, revealing the likelihood of students moving from one state to another [12, 13]. Centrality measures identify influential states within the transition process, showing which self-regulatory patterns serve as hubs in the network. This approach aligns with complex systems principles by emphasizing dynamic reconfiguration over static variable associations [3].

Self-regulated learning (SRL) has been extensively studied as a dynamic, multifaceted process. SRL involves planning, monitoring, and controlling cognitive, motivational, and behavioral aspects of learning [7]. Traditional approaches to measuring SRL have relied heavily on self-report questionnaires [14], which present

challenges when implemented repeatedly during longitudinal studies. Therefore, methods that reflect the dynamics are needed to capture these longitudinal dynamics.

Among these dynamics, transitions between SRL states during a course is a key interest in SRL research. Understanding these dynamics provides insights into patterns of change and potential educational interventions [15].

3 Methods

3.1 Data Collection and Participants

The study used longitudinal survey data from 165 students in an online project management course at a Finnish higher education institution. The semester-long, asynchronous course allowed individual progress. Students completed the survey at the start and before each assignment submission, capturing dynamic variations in their self-regulatory behaviors in natural online learning environments [16]. The university approved the research, and students provided informed consent for participation.

3.2 Survey Instrument

In developing our measurement approach, we recognized the challenge inherent in longitudinal SRL research. While comprehensive SRL inventories exist with numerous items [17], these extensive instruments become problematic when implemented repeatedly during a course [18]. Research has established that shorter instruments are preferable for tracking developmental trajectories in longitudinal designs [19], particularly when monitoring competency development over time.

Our approach used Ecological Momentary Assessment (EMA), specifically an EMA questionnaire (EMA Q), delivering brief measures to students at multiple timepoints throughout the course. EMA captures students' self-regulatory behaviors in real-time within their natural learning environments, minimizing recall bias and maximizing ecological validity and provides more accurate data than retrospective assessments, which often contain significant recall biases [20]. Research shows concise, single-item constructs can maintain adequate reliability compared to longer alternatives [21].

The instrument comprised 11 targeted statements addressing key SRL dimensions, carefully selected from established validated measures [14, 22, 23]. These items were specifically chosen to align with the course structure, pedagogical framework, and implementation logistics. Following established protocols in longitudinal SRL research [24, 25], we prioritized brevity and theoretical coverage over comprehensive psychometric validation for this specific configuration. All variables were standardized using the normalization method to ensure comparability across different scales and response patterns.

3.3 Data Analysis

Our analytical approach is based on complex dynamic systems theory, which views learning as emergent phenomena with non-linear interactions, attractor states, and phase transitions [26]. Complex systems exhibit self-organization, sensitivity to initial conditions, and stable patterns arising from dynamic interactions [27]. This framework guided our two-stage methodology: cluster analysis identifies self-regulatory profiles as potential patterns in the data, while transition network analysis captures dynamic movements between these patterns. We operationalize CDS concepts by treating cluster stability as statistical patterns analogous to attractor states and transition probabilities as indicators of system dynamics[28]. This method aligns with recommendations for

studying complex educational processes by modeling structural and temporal dimensions simultaneously [6, 28].

To answer **RQ1,** we identified distinct student profiles based on self-regulatory patterns using a Mixture of Experts (MoE) clustering approach with the MoEClust R package [29]. We chose this method over traditional clustering because it better handles complex, multivariate data and accounts for uncertainty in cluster membership [30]. Model selection compared Bayesian Information Criterion (BIC) values across models with 1-6 clusters, as recommended by [31]. The optimal solution was a five-cluster model with an ellipsoidal, equal shape (VEV) configuration, balancing complexity with explanatory power [32]. This decision was supported by statistical criteria and the theoretical interpretability of clusters. The resulting profiles were labeled based on their characteristic patterns across SRL dimensions, drawing on theoretical frameworks of self-regulation [7, 14]. These labels represent interpretive categorizations of observed patterns rather than predetermined psychological types. Clustering quality was evaluated using entropy metrics and average posterior probabilities [33].

To answer **RQ2**, we analyzed students' transitions between self-regulatory states throughout the course using sequence analysis with the TraMineR [34] and TNA with tna R packages [12, 13]. TNA captures the longitudinal and relational dynamics in longitudinal data and models the transitions between states [12, 13]. TNA requires sequence data which we prepared using the student's cluster membership data over time following the method of [25].

Centrality measures (InStrength and OutStrength) identify the network's central and influential states [35], revealing key self-regulatory hubs in the transition process. InStrength emphasizes the most important learning [12]. In practical terms, high InStrength indicates a profile that many students transition toward, while high OutStrength indicates a profile from which students frequently transition to other states (temporary). To ensure robustness, bootstrapping was conducted with a consistency range of 0.75 to 1.25 [12, 36], providing confidence intervals for transition probabilities and identifying the most stable network transitions [12, 37].

4 Results

In response to **RQ1**, our mixture of experts (MoE) cluster analysis yielded five distinct self-regulatory profiles. The five-cluster model with an ellipsoidal, equal shape (VEV) configuration demonstrated optimal fit based on Bayesian Information Criterion (BIC = -44538.6) and high entropy metrics (0.896). The average posterior probabilities ranged from 0.844 to 1.000 across the five clusters, indicating strong separation and reliable assignment of observations to clusters.

The profiles revealed distinct patterns across 11 self-regulatory variables from our survey. We labeled these as: Hopeful, Discouraged, Realistic, Self-Confident, and Deregulators. The distribution showed the Hopeful profile was most prevalent (n=440, 37.9%), followed by Discouraged (n=343, 29.5%), Deregulators (n=204, 17.6%), Realistic (n=139, 12.0%), and Self-Confident (n=38, 3.3%). These profiles represent momentary self-regulatory behaviors measured through the EMA Q, offering insights into how students manage learning in real-time instead of relying on retrospective self-reports.

Fig. 1 shows the mean scores of each self-regulatory variable across the five profiles. The Hopeful profile had high scores in all dimensions, especially in criteria establishment, deadline adherence, and strategic planning. In contrast, the Discouraged profile had below-average scores, particularly in self-assessment and help-seeking behaviors. The Realistic profile had moderate scores, notably in time management and adherence to standards. The Self-Confident profile exhibited high self-assessment scores but lower scores in help-seeking and schedule adherence. Finally, the Deregulators profile had low scores across all dimensions, reflecting minimal engagement with self-regulatory strategies.

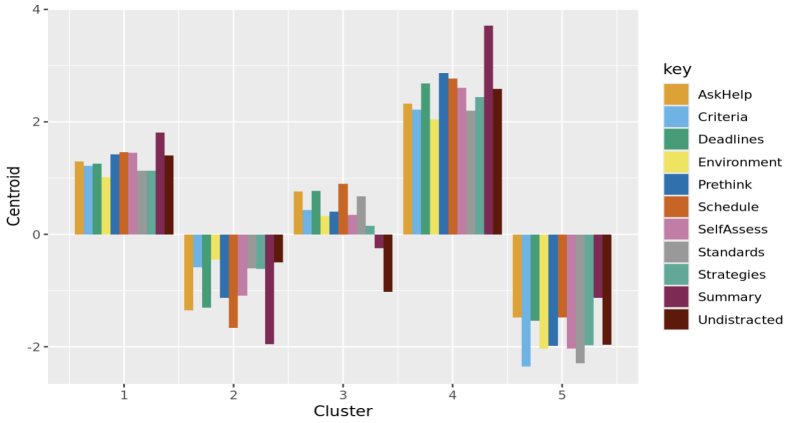


Fig. 1. Cluster profiles and their standardized values for different instrument item responses.

For **RQ2**, our analyses revealed distinct patterns in how students transitioned between self-regulatory states. Some maintained consistent profiles, while many shifted states, highlighting the dynamic nature of self-regulation in online learning.

Analysis of initial probabilities showed students most often started in the Discouraged state (40.6%), followed by Hopeful (22.4%), Realistic (19.4%), Deregulators (13.9%), and Self-Confident (3.6%). This pattern suggests many students enter online courses with lower self-regulatory confidence, though we note these represent self-reported states at course beginning rather than stable traits. The transition probability matrix revealed the likelihood of students moving between different states. The probability of remaining in the same state was notably high across profiles: Hopeful (0.87), Deregulators (0.83), Discouraged (0.76), Self-Confident (0.76), and Realistic (0.43). These high self-loop probabilities indicate statistical stability in students' reported regulatory patterns.

Observed transitions revealed key patterns. Students in the Realistic profile showed the highest likelihood of transitioning to the Hopeful state (0.33), suggesting movement toward more positive self-regulatory patterns. Students in the Discouraged profile frequently transitioned to the Deregulators state (0.10), indicating movement toward lower regulatory engagement. The Self-Confident profile showed few transitions except to Hopeful (0.21), suggesting a more isolated pattern.

The filtered network (Fig. 2, left) displays transitions with probabilities over 0.05, highlighting common transitions between regulatory states. This visualization emphasizes the Hopeful and Discouraged profiles as key hubs for student transitions.

Initial probabilities (outer circle of nodes) are highest for discouraged and hopeful states, with self-loops indicating a tendency to stay in the same state. When transitions occur, they likely shift from realistic to hopeful (0.33) or from Self-Confident to Hopeful (0.21). Bootstrap analysis with 1000 resamples confirmed the robustness of our model (Fig. 2, right). The consistency-filtered network retained most major transition pathways from the original analysis, indicating stability across different data subsets.

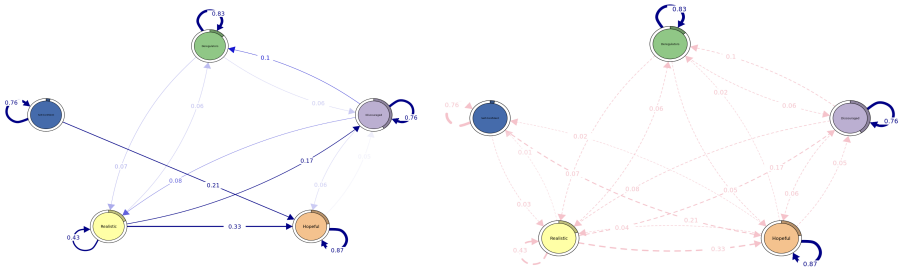


Fig. 2. Transition network between different states (nodes) and the transition probabilities between states (edges). Left: Filtered transition network shows only transitions with probabilities above 0.05. Right: Bootstrap-validated transition network showing the consistency of transition patterns, with only robust transitions retained.

Analysis of centrality measures (Fig. 3) quantified the importance of different profiles in the transition network. The InStrength measure, which represents the likelihood of students transitioning into a state, was highest for the Hopeful profile, followed by Discouraged. The OutStrength measure, indicating transitions from one state to others, was highest for the Realistic profile, confirming it as the most volatile state and showing that Hopeful was the most recurrent state.

Overall, our transition network analysis revealed both stability and meaningful change in students' self-regulatory approaches during the online course. The identified transition patterns suggest observable pathways in self-regulation, with particular profiles serving as key transitional states that may represent important intervention points for educational support.

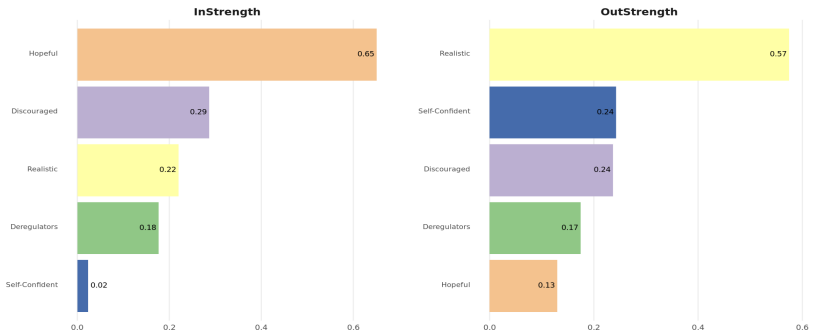


Fig. 3. Centrality measures (InStrength on the left and OutStrength on the right) for the transition network.

5 Discussion

Our findings support viewing self-regulated learning as a dynamic system. The emergence of five self-regulatory profiles aligns with the concept of heterogeneity and individual differences which emphasizes that students have different profiles of regulation. The “loops” in the transition networks are consistent with the concept of sensitivity to initial states and stability over time—patterns consistent with the complex systems notion of attractor states [3]. High probabilities of remaining in the same state (especially for Hopeful and Deregulator profiles) suggest these patterns exhibit strong stability. In practical terms, this means students reporting these regulatory patterns at one timepoint are highly likely to report similar patterns at subsequent timepoints. This stability pattern has implications for instance, that changes won’t happen without investment of effort and time to help students change their attitudes and behaviors.

Our findings on self-regulatory profiles align with Zimmerman's [7] cyclical model, suggesting established regulatory patterns persist through reinforcement mechanisms. The high self-loop probabilities observed (in the Hopeful and Deregulators profiles) echo Stamovlasis's [5] discussion of stable patterns in educational systems. From a complex systems perspective [3], these stable states represent preferred configurations the system naturally maintains.

The 40.6% of students starting in the Discouraged state underscores the significance of initial conditions in complex systems. This prevalence pattern aligns with [36] observation that online learning often overwhelms students' regulatory capabilities. This pattern suggests that early instructional support may be particularly important, as initial states appear to influence subsequent patterns.

The observed transition patterns, particularly from Realistic to Hopeful states, align with [15] findings on self-regulatory capabilities development. This transition may represent a reorganization of regulatory behaviors consistent with complex systems theory [38]. Conversely, the transition from Discouraged to Deregulators states represents movement toward lower regulatory engagement, as identified by [39] in their analysis of self-regulatory dynamics.

Our network centrality findings showed that high InStrength of the Hopeful profile makes it a primary hub in the regulatory network, attracting transitions from various states. This centrality pattern indicates that the Hopeful profile represents a common destination state—many students who change their regulatory patterns tend to move toward this configuration[40].

This study has several important limitations. First, we did not conduct comprehensive psychometric validation of our 11-item instrument in its specific configuration, instead relying on the validation of items in their original instruments and following established precedents in longitudinal SRL research. While this approach is common in repeated-measures studies to minimize participant burden, it limits our ability to make strong claims about construct validity. Second, data collection timing (before assignment submissions) may introduce bias, as students' regulatory states might differ during active work periods versus submission points. Third, our single-institution, single-course design limits generalizability. Fourth, the small number of students in some profiles (particularly Self-Confident) constrains our analysis of their transition patterns. Finally, while we model transitions between profiles, the profiles themselves are treated as static categories rather than potentially evolving constructs.

This study reveals promising avenues for future research. First, intervention studies that examine whether targeted support can facilitate transitions from less to more productive regulatory patterns are crucial. Second, expanding the complex dynamic systems approach with additional parameters would enhance understanding of self-regulatory dynamics. Future research could explore how contextual factors (e.g., course design), personal characteristics (e.g., goal orientations), and social dimensions (e.g., peer interactions) influence regulatory state transitions.

6 Conclusions

This study highlights the value of viewing self-regulated learning (SRL) as a complex dynamic system when analyzing student behaviors in online environments. When identifying distinct self-regulatory profiles and transition patterns, we reveal both stability and change in students' learning regulation throughout an online course. Most importantly, our findings show that students are unlikely to change their self-regulatory behaviors independently. When changes do occur, they often revert to their initial states, emphasizing the need for timely and early interventions. Designing effective support strategies requires understanding that change demands significant effort.

Methodologically, this study contributes by applying TNA to SRL, allowing for the simultaneous examination of both structural patterns and temporal dynamics. Identifying statistical stability patterns, transition pathways, and key hubs within the regulatory network, this work offers a framework for understanding the complexities of self-regulation, which can inform targeted interventions in online education.

References

1. Broadbent, J., Poon, W.L.: Self-regulated learning strategies & academic achievement in online higher education learning environments: A systematic review. Internet High. Educ. 27, 1–13 (2015).
2. Jacobson, M.J., Kapur, M., Reimann, P.: Conceptualizing debates in learning and educational research: Toward a complex systems conceptual framework of learning. Educ. Psychol. 51, 210–218 (2016).
3. Koopmans, M., Stamovlasis, D. eds: Complex dynamical systems in education. Springer International Publishing, Cham, Switzerland (2018).
4. Saqr, M., Dever, D., López-Pernas, S., Gernigon, C., Marchand, G., Kaplan, A.: Complex dynamic systems in education: Beyond the static, the linear and the causal reductionism, https://lamethods.github.io/ch12-cds.html, (2024). https://doi.org/10.48550/ARXIV.2501.10386.
5. Stamovlasis, D.: Nonlinear dynamical interaction patterns in collaborative groups: Discourse analysis with orbital decomposition. In: Complex Dynamical Systems in Education. pp. 273–297. Springer International Publishing, Cham (2016).
6. Hilpert, J.C., Marchand, G.C.: Complex systems research in educational psychology: Aligning theory and method. Educ. Psychol. 53, 185–202 (2018).
7. Zimmerman, B.J.: Investigating self-regulation and motivation: Historical background, methodological developments, and future prospects. Am. Educ. Res. J. 45, 166–183 (2008).
8. Gasevic, D., Jovanovic, J., Pardo, A., Dawson, S.: Detecting learning strategies with analytics: Links with self-reported measures and academic performance. J. Learn. Anal. 4, 113–128 (2017).

9. López-Pernas, S., Saqr, M.: The why, the how and the when of educational process mining in R. In: Learning Analytics Methods and Tutorials. pp. 467–488. Springer Nature Switzerland, Cham (2024).
10. López-Pernas, S., Saqr, M., Viberg, O.: Putting it all together: Combining learning analytics methods and data sources to understand students' approaches to learning programming. Sustainability. 13, 4825 (2021).
11. Reimann, P., Markauskaite, L., Bannert, M.: e-Research and learning theory: What do sequence and process mining methods contribute? Br. J. Educ. Technol. 45, 528–540 (2014).
12. Saqr, M., López-Pernas, S., Törmänen, T., Kaliisa, R., Misiejuk, K., Tikka, S.: Transition network analysis: A novel framework for modeling, visualizing, and identifying the temporal patterns of learners and learning processes. In: Proceedings of the 15th International Learning Analytics and Knowledge Conference. pp. 351–361. ACM, New York, NY, USA (2025).
13. Tikka, S., López-Pernas, S., Saqr, M.: Tna: An R package for transition network analysis. Appl. Psychol. Meas. 01466216251348840 (2025).
14. Barnard, L., Lan, W.Y., To, Y.M., Paton, V.O., Lai, S.-L.: Measuring self-regulation in online and blended learning environments. Internet High. Educ. 12, 1–6 (2009).
15. Hadwin, A.F., Nesbit, J.C., Jamieson-Noel, D., Code, J., Winne, P.H.: Examining trace data to explore self-regulated learning. Metacogn. Learn. 2, 107–124 (2007).
16. Russell, M.A., Gajos, J.M.: Annual Research Review: Ecological momentary assessment studies in child psychology and psychiatry. J. Child Psychol. Psychiatry. 61, 376–394 (2020).
17. Toering, T., Elferink-Gemser, M.T., Jonker, L., van Heuvelen, M.J.G., Visscher, C.: Measuring self-regulation in a learning context: Reliability and validity of the Self-Regulation of Learning Self-Report Scale (SRL-SRS). Int. J. Sport Exerc. Psychol. 10, 24–38 (2012).
18. Choi, B.C.K., Pak, A.W.P.: A catalog of biases in questionnaires. Prev. Chronic Dis. 2, A13 (2005).
19. Wolters, C.A., Won, S.: Validity and the use of self-report questionnaires to assess self-regulated learning. In: Handbook of Self-Regulation of Learning and Performance. pp. 307–322. Routledge (2017).
20. Shiffman, S., Stone, A.A., Hufford, M.R.: Ecological momentary assessment. Annu. Rev. Clin. Psychol. 4, 1–32 (2008).
21. Gogol, K., Brunner, M., Goetz, T., Martin, R., Ugen, S., Keller, U., Fischbach, A., Preckel, F.: "My Questionnaire is Too Long!" The assessments of motivational-affective constructs with three-item and single-item measures. Contemp. Educ. Psychol. 39, 188–205 (2014).
22. Jansen, R.S., van Leeuwen, A., Janssen, J., Kester, L., Kalz, M.: Validation of the self-regulated online learning questionnaire. J. Comput. High. Educ. 29, 6–27 (2017).
23. Broadbent, J., Panadero, E., Lodge, J.M., Fuller-Tyszkiewicz, M.: The self-regulation for learning online (SRL-O) questionnaire. Metacogn. Learn. 18, 135–163 (2022).
24. Heikkinen, S., Saqr, M., Malmberg, J. & Tedre, M.: The interplay of engagement and learning regulation in online learning. In: Proceedings of TEEM 2024. Technological Ecosystems for Enhancing Multiculturality 2024. Springer, Singapore, Singapore.
25. Heikkinen, S., Saqr, M., Malmberg, J., Tedre, M.: A longitudinal study of interplay between student engagement and self-regulation. Int. J. Educ. Technol. High. Educ. 22, (2025). https://doi.org/10.1186/s41239-025-00523-3.
26. van Dijk, M., Lowie, W., Smit, N., Verspoor, M., van Geert, P.: Complex dynamic systems theory as a foundation for process-oriented research on second language development. Second Lang. Res. (2024). https://doi.org/10.1177/02676583241246739.
27. Kaplan, A., Garner, J.K.: A complex dynamic systems perspective on identity and its

development: The dynamic systems model of role identity. Dev. Psychol. 53, 2036–2051 (2017).
28. López-Pernas, S., Saqr, M.: From Variables to States to Trajectories (VaSSTra): A Method for Modelling the Longitudinal Dynamics of Learning and Behaviour. In: Proceedings TEEM 2022: Tenth International Conference on Technological Ecosystems for Enhancing Multiculturality. pp. 1169–1178. Springer Nature Singapore (2023).
29. Murphy, K., Murphy, T.B.: Gaussian parsimonious clustering models with covariates and a noise component. Adv. Data Anal. Classif. 14, 293–325 (2020).
30. Vermunt, J.K., Magidson, J.: Latent Class Cluster Analysis. In: Hagenaars, J.A. and McCutcheon, A.L. (eds.) Applied Latent Class Analysis. pp. 89–106. Cambridge University Press, Cambridge (2002).
31. Nylund, K.L., Asparouhov, T., Muthén, B.O.: Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. Struct. Equ. Modeling. 14, 535–569 (2007).
32. Scrucca, L., Fop, M., Murphy, T.B., Raftery, A.E.: Mclust 5: Clustering, classification and density estimation using Gaussian finite mixture models. R J. 8, 289–317 (2016).
33. Celeux, G., Soromenho, G.: An entropy criterion for assessing the number of clusters in a mixture model. J. Classif. 13, 195–212 (1996).
34. Gabadinho, A., Ritschard, G., Müller, N.S., Studer, M.: Analyzing and visualizing state sequences inRwithTraMineR. J. Stat. Softw. 40, (2011). https://doi.org/10.18637/jss.v040.i04.
35. Borgatti, S.P.: Centrality and network flow. Soc. Networks. 27, 55–71 (2005).
36. Efron, B., Tibshirani, R.J.: An introduction to the bootstrap. Chapman & Hall/CRC, Philadelphia, PA (2011).
37. Costenbader, E., Valente, T.W.: The stability of centrality measures when networks are sampled. Soc. Networks. 25, 283–307 (2003).
38. Fisher, A.J., Medaglia, J.D., Jeronimus, B.F.: Lack of group-to-individual generalizability is a threat to human subjects research. Proc. Natl. Acad. Sci. U. S. A. 115, E6106–E6115 (2018).
39. Saqr, M., López-Pernas, S.: Mapping the self in self-regulation using complex dynamic systems approach. Br. J. Educ. Technol. 55, 1376–1397 (2024).
40. Saqr, M., Peeters, W.: Temporal networks in collaborative learning: A case study. Br. J. Educ. Technol. 53, 1283–1303 (2022).